

- Question 1 *Compute the LU factorization of the following matrix*

$$A = \begin{bmatrix} 1 & 2 & 7 \\ 3 & 5 & -1 \\ 6 & 1 & 4 \end{bmatrix}$$

in the following two ways:

1. *Gaussian elimination*

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1 A = \begin{bmatrix} 1 & 2 & 7 \\ 0 & -1 & -22 \\ 6 & 1 & 4 \end{bmatrix}$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix}$$

$$E_2 E_1 A = \begin{bmatrix} 1 & 2 & 7 \\ 0 & -1 & -22 \\ 0 & -11 & -38 \end{bmatrix}$$

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -11 & 1 \end{bmatrix}$$

$$E_3 E_2 E_1 A = \begin{bmatrix} 1 & 2 & 7 \\ 0 & -1 & -22 \\ 0 & 0 & 204 \end{bmatrix}$$

$$= U$$

$$\begin{aligned}
L &= (E_3 E_2 E_1)^{-1} \\
&= E_1^{-1} E_2^{-1} E_3^{-1} \\
&= E_1^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 6 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 11 & 1 \end{bmatrix} \\
&= E_1^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 6 & 11 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 6 & 11 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 6 & 11 & 1 \end{bmatrix}
\end{aligned}$$

2. Doolittle's method

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} A = LU$$

$$\begin{aligned}
u_{11} &= 1 \\
3 &= u_{11}l_{21} \\
3 &= l_{21} \\
6 &= u_{11}l_{31} \\
6 &= l_{31} \\
u_{12} &= 2 \\
5 &= l_{21}u_{12} + u_{22} \\
5 &= 6 + u_{22} \\
-1 &= u_{22} \\
1 &= u_{12}l_{31} + u_{22}l_{32} \\
1 &= 2 \cdot 6 - l_{32} \\
-11 &= -l_{32} \\
11 &= l_{32} \\
u_{13} &= 7 \\
-1 &= u_{13}l_{21} + u_{23} \\
-1 &= 7 \cdot 3 + u_{23} \\
-22 &= u_{23} \\
4 &= u_{13}l_{31} + u_{23}l_{32} + u_{33} \\
4 &= 7 \cdot 6 - 22 \cdot 11 + u_{33} \\
204 &= u_{33}
\end{aligned}$$

Therefore

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 6 & 11 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 7 \\ 0 & -1 & -22 \\ 0 & 0 & 204 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

- Question 2 *Solve the triadiagonal system by first computing an LU factorization of the coefficient matrix and then using the LU factorization to solve the following system:*

$$\begin{aligned}
2x_1 + x_2 &= 3 \\
-x_1 + 2x_2 + x_3 &= 2 \\
-x_2 + 2x_3 + x_4 &= 2 \\
-x_3 + 2x_4 + x_5 &= 2 \\
-x_4 + 2x_5 &= 1
\end{aligned}$$

We're trying to find $(x_1, x_2, x_3, x_4, x_5)$. Note by the tri-diagonal we need to establish

the matrix corresponding to this system of equations. We have the following:

$$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ -1 & 2 & 1 & 0 & 0 \\ 0 & -1 & 2 & 1 & 0 \\ 0 & 0 & -1 & 2 & 1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} b_1 & c_1 & 0 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 0 \\ 0 & a_3 & b_3 & c_3 & 0 \\ 0 & 0 & a_4 & b_4 & c_4 \\ 0 & 0 & 0 & a_5 & b_5 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 2 \\ 2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \end{bmatrix}$$

Note by the tridiagonal algorithm we have that

$$c'_1 = \frac{1}{2}, c'_2 = \frac{1}{2 + \frac{1}{2}} = \frac{2}{5}, c'_3 = \frac{1}{2 + \frac{2}{5}} = \frac{5}{12}, c'_4 = \frac{1}{2 + \frac{5}{12}} = \frac{12}{29}$$

$$d'_1 = \frac{3}{2}, d'_2 = \frac{2 + \frac{3}{2}}{2 + \frac{1}{2}} = \frac{7}{5}, d'_3 = \frac{2 + \frac{7}{5}}{2 + \frac{2}{5}} = \frac{17}{12}, d'_4 = \frac{2 + \frac{17}{12}}{2 + \frac{5}{12}} = \frac{41}{29}, d'_5 = \frac{1 + \frac{41}{29}}{2 + \frac{12}{29}} = \frac{70}{70} = 1$$

$$x_5 = 1, x_4 = \frac{41}{29} - \frac{12}{29} = 1, x_3 = \frac{17}{12} - \frac{5}{12} = 1, x_2 = \frac{7}{5} - \frac{2}{5} = 1, x_1 = \frac{3}{2} - \frac{1}{2} = 1$$

Therefore $(x_1, x_2, x_3, x_4, x_5) = (1, 1, 1, 1, 1)$

- Question 3 Define the matrix $A_n \in \mathbb{R}^{n \times n}$ by

$$A_n = \begin{bmatrix} 1 & -1 & -1 & \cdots & -1 \\ 0 & 1 & -1 & \cdots & -1 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -1 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

1. Show that the entry of the inverse of A_n at row i and column j has value:

$$(a_{ij})^{-1} = \begin{cases} 0, & \text{if } i > j \\ 1, & \text{if } i = j \\ 2^{j-i-1}, & \text{if } i < j \end{cases}$$

We will prove this by induction. Note for the base case, $n = 1$, that $A_1 = [1]$, and therefore $A_1^{-1} = [1]$, which is just element $a_1 1 = 1$. Thus the base case holds. The induction hypothesis is as follows: let $k \in \mathbb{N}$, if $k \leq n$ then we have

for the matrix A_k that the inverse corresponds to $(A_k^{-1})_{ij} = \begin{cases} 0, & \text{if } i > j \\ 1, & \text{if } i = j \\ 2^{j-i-1}, & \text{if } i < j \end{cases}$

Now we want to show this is the case for A_{n+1} . Note that for A_n we have the well defined inverse A_n^{-1} by the induction hypothesis. Note furthermore that the

$$\text{matrix } A_n^{-1'} = \left[\begin{array}{c|c} A_n^{-1} & \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} \\ \hline 0 \cdots 0 & 1 \end{array} \right]$$

2. *What is the condition number of A_n ? Is the matrix ill-conditioned?*

$$\|A\|_\infty = \sum_{i=1}^n 1 = n$$